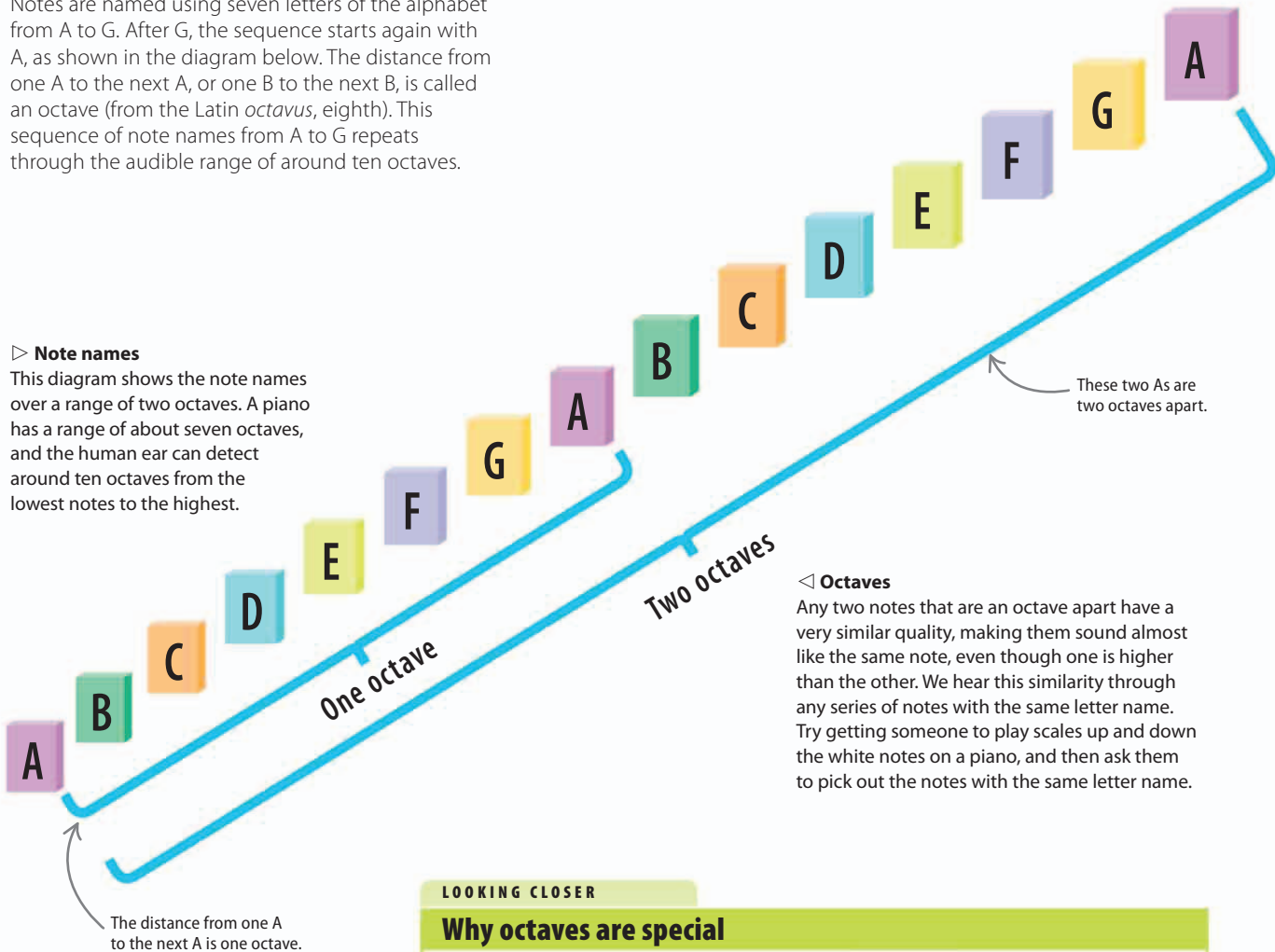


Notes and octaves

Notes are named using seven letters of the alphabet from A to G. After G, the sequence starts again with A, as shown in the diagram below. The distance from one A to the next A, or one B to the next B, is called an octave (from the Latin *octavus*, eighth). This sequence of note names from A to G repeats through the audible range of around ten octaves.

▷ Note names

This diagram shows the note names over a range of two octaves. A piano has a range of about seven octaves, and the human ear can detect around ten octaves from the lowest notes to the highest.



◁ Octaves

Any two notes that are an octave apart have a very similar quality, making them sound almost like the same note, even though one is higher than the other. We hear this similarity through any series of notes with the same letter name. Try getting someone to play scales up and down the white notes on a piano, and then ask them to pick out the notes with the same letter name.

Playing **octaves with one hand** is a characteristic of **ragtime piano** playing and many **other styles**.

LOOKING CLOSER

Why octaves are special

There is a scientific reason why two notes an octave apart sound similar, even though one is higher-pitched than the other. This is because the higher note vibrates exactly twice as fast as the lower one.

▷ Sound waves

This diagram represents two sound waves. The upper one is vibrating twice as fast as the lower one, so the sound we hear from it is one octave higher.

